Coursework Project

Description

The Morgana discovery is a gas field located in ca. 40 m water depth in the Tethys Sea, approximately 25 km from the coast. The sea temperature varies between 8 °C and 16 °C; Metocean conditions are generally benign. The first exploration well 15/9-A was drilled in 2006 but was water-bearing. Well 15/9-B, drilled by Tyche Petroleum in 2010 on a 2D seismic line, discovered gas-bearing sandstones in a reservoir of Bajocian age. A subsequent appraisal well, 15/9-C, penetrated the crest of the structure and flowed at a maximum rate of 1 Msm^3/d on test. Figures 1 and 2 show the top and base Bajocian structure maps, and the location of the 3 wells drilled to date. Only 2D seismic is currently available. Figure 3 shows the correlation between the 3 wells.

Limited test data for the 15/9-C flow test is provided in the file `coursework_data.xlsx` that accompanies this exercise. This file contains the following information:

- petrophysical sums and averages for a range of petrophysical cut-offs
- MDT data
- capillary pressure data
- PVT data
- well test data
- VFP data
- production data from the nearby Merlin field

A gas sample during the 15/9-C well test had an average specific gravity of 0.65 relative to air (air density is 1.293 kg/m^3) and also showed 50 ppm H_2S and 0.1 mol% CO_2. Further PVT analysis indicated a CGR of ca. 400 sm^3/Msm^3, which suggests that reservoir fluid contains a significant condensate volume. The reservoir water is relatively fresh and has an equivalent gradient of 0.100 bar/m. Geomechanical tests show that the reservoir rock is moderately well-cemented, and require the initial drawdown to be limited to 100 bar to reduce the risk of sand production. The reservoir temperature is 90 °C at a datum depth of 3000 m tvss. The geothermal gradient is 25 °C/km. Standard conditions are 15 °C and 1 bara.

The current outline development concept assumes gas sales to the domestic market, with an initial minimum flowing tubing head pressure of 100 bara; this can be reduced in stages to 60 bara and 20 bara by reconfiguring the compression ssytem. The draft gas sales contract requires a 10 year plateau period (wellhead gas) and government regulations define a maximum decline rate post plateau of 30% per annum. The minimum economic gas rate is 0.20 Msm^3/d.

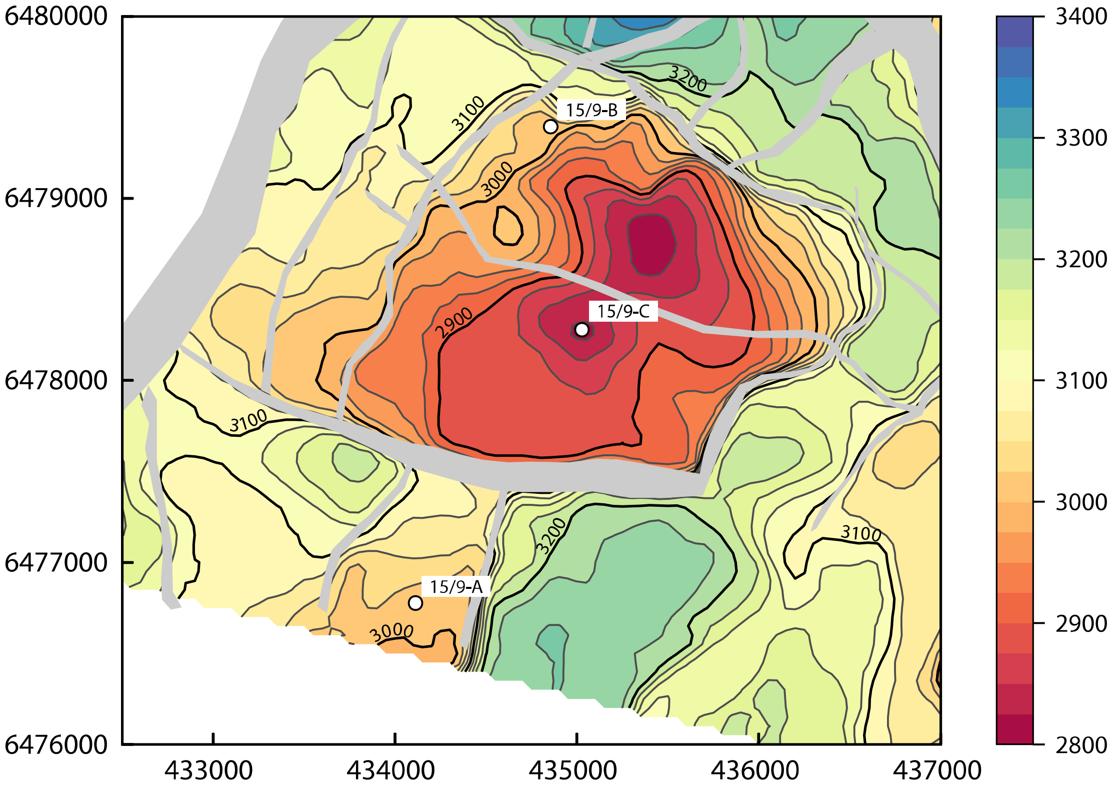

Figure 1: Top Bajocian structure map

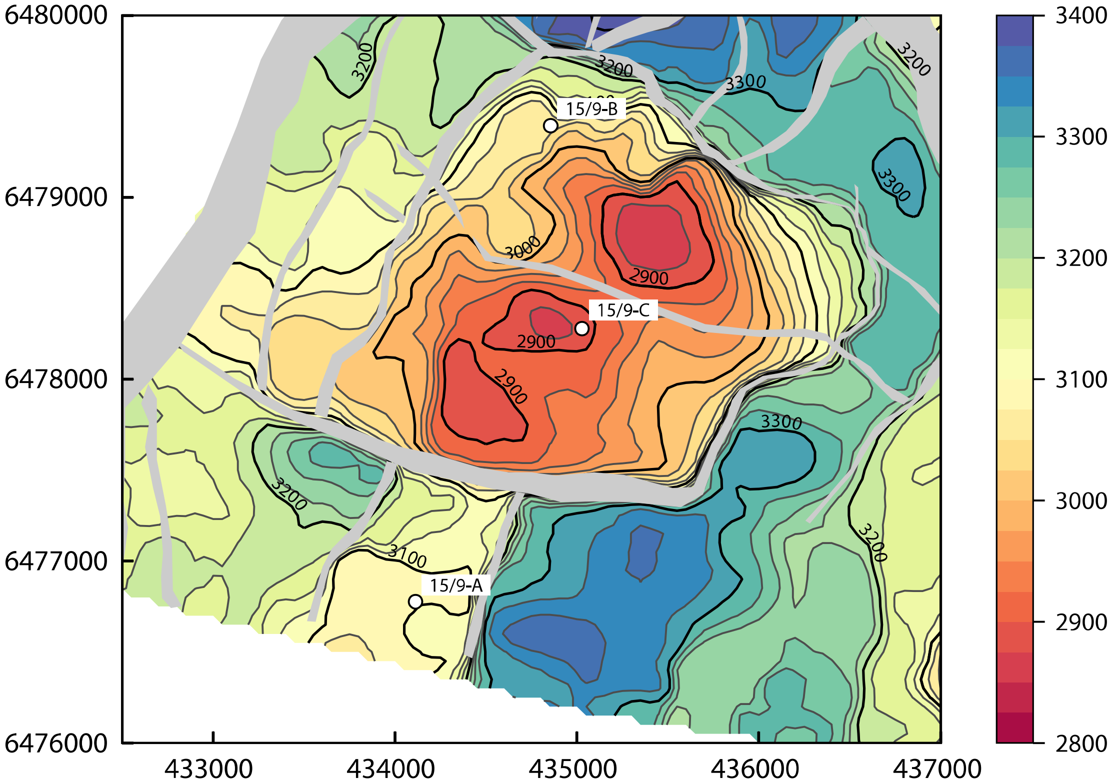

Figure 2: Base Bajocian structure map

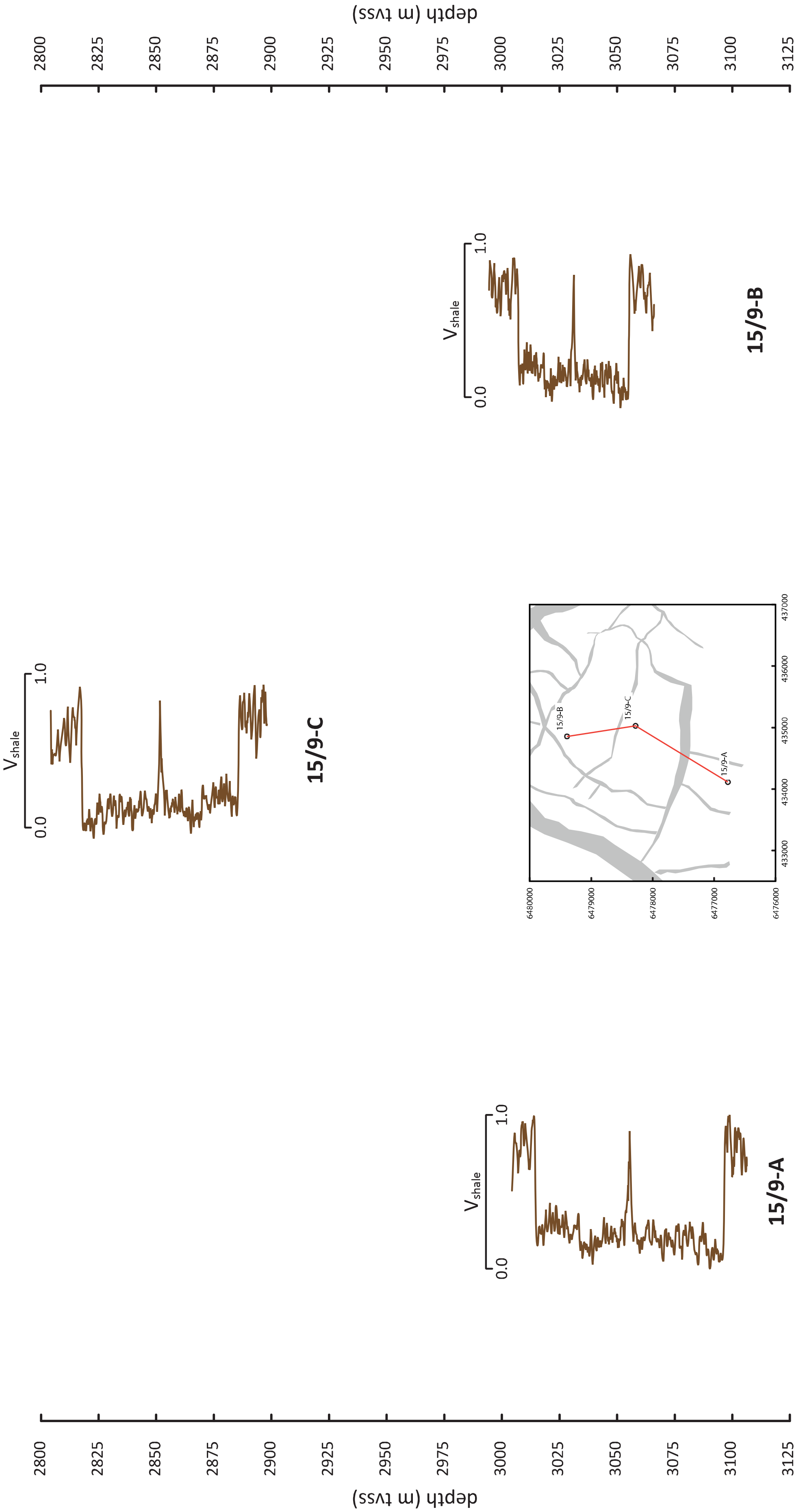

Figure 3: Correlation section; Bajocian reservoir

Tyche Petroleum are considering selling this field as part of their current portfolio rationalisation process. Your company is considering acquiring the Tyche interest and have asked you to prepare a short briefing note to define the mid case production profile and an associated development plan for the field (less than 10 pages text, excluding Appendices). Your briefing note should address the following issues:

(a) *What is the range of GIIP?*

You should calculate a range of GIIPs volumetrically using the data provided. Outline the assumptions and calculations that under-lie your GIIP estimates, including the approach used to estimate uncertainty in GIIP. These estimates should be expressed as Low/Mid/High (LMH) (eg.1P/2P/3P or P90/P50/P10) estimates.

(b) *What is the LMH range of RF and UR?*

The file `coursework_data.xlsx` contains inflow and outflow performance data for a typical Morgana gas production well, and production data from the adjacent Merlin field. Use this data to estimate a range of recovery factors (RF) and ultimate recovery (UR).

(c) *What is the initial well production rate?*

The test data from well 15/9-C provides sufficient information to estimate the average initial production rate from a development well. Is there a range?

(d) *How many wells are required?*

The well count should maximise the production rates expected from the field while honouring the post-plateau decline rate constraint given above. You should indicate the location of the development wells on the top structure map and describe briefly the rationale for choosing these locations.

(e) *What is the range of production profiles?*

You should provide the annual average LMH production rate for the field so that the economist can estimate the Net Present Value (NPV) of the field. You should cut-off the profile at the economic rate provided and estimate the reserves in the Morgana field. Your profile should also include an estimate of the liquid (condensate) production rate and should account for shrinkage. You should also indicate the time at which compression will be required.

(f) *Are there any potential production problems?*

Your management would also like to know if there are any potential production problems associated with the development and their mitigations.

(g) *What are the key subsurface uncertainties associated with the development of this reservoir?*

You should summarise these uncertainties and indicate what additional data (if any) is required to address them. Is further appraisal needed?

(h) *What would be an appropriate development concept?*

Suggest the type of development concept that might be considered for development of the Morgana field. How will you access wells when the field is in production to carry out routine surveillance (eg downhole pressure surveys)?